How accurately does the ask surface read a regulatory review?

Measured over 14 documents and 55 questions. Generated 2026-08-14 05:51 UTC from commit 6b626cf.

```
model (answering) . . gemini-3.5-flash
retriever . . . . . . BM25 over pages, Porter stemming, phrase expansion, concept map
(services/api/terms.ts)
temperature . . . . . 0, and the retriever is deterministic, so variation across runs is the model
alone
inputs. . . . . . . . data/retrieval-eval.json (fixture) · results/retrieval-eval.json · results/ask-
eval.json
reproduce . . . . . . npm run retrieval:eval  |  npm run ask:eval -- --repeats=3  |  npm run report
```

1. What this measures, and what it does not

The surface takes a question about one regulatory review, retrieves 16 pages, and answers from those pages while naming the page each claim rests on. Two things can go wrong independently, so they are measured separately: the retriever can fail to surface the page holding the answer, and the model can misread a page it was given. Every number below separates them.

> **The honest limits, stated before the results rather than after.** The gold pages were located by regular expression and then read, which biases every comparison toward a lexical retriever. The answer check asks whether the answer *states* the fact its gold quote carries — it catches an answer that missed the number, not one that described it wrongly. Graded correctness needs a judge model that is not the answering model, reported with a human-agreement figure beside it; that is not in this report. And the gold set names pages that are *sufficient*, never all the pages that would do, so citation overlap outside the gold set is not error.

2. The documents

14 readable documents, each an FDA multi-disciplinary review or an EMA assessment report of an approved drug, fetched by NDA number from accessdata.fda.gov and screened with `data/prep/measure_pdf.py` before use. The set varies on purpose: a boxed hepatotoxicity warning, liver findings without one, approved drugs whose labels carry no liver warning at all, and one 505(b)(2) with no new nonclinical studies. A set of only hepatotoxic drugs measures willingness to say danger and nothing else.

document	pages	characters	MB	sha256 (first 16)
Turalio (pexidartinib) - FDA NDA 211810 multi-disciplinary review	264	528,734	6.1	aa5faedfe4e0b782
Imaavy (nipocalimab) - EMA CHMP assessment report	178	487,124	8.5	145b2783a91a21cd
Slynd (drospirenone) - FDA NDA 211367 multi-disciplinary review	132	259,928	6.3	dcb3b327556d2e16
Lumakras (sotorasib) - FDA NDA 214665 multi-disciplinary review	269	470,954	5.6	59f6cad47a05cb44
Retevmo (selpercatinib) - FDA NDA 213246 multi-disciplinary review	398	807,847	18.1	b995955bf303f6f4

Trikafta (elexacaftor/tezacaftor/ivacaftor) - FDA NDA 212273 multi-disciplinary review	245	550,105	6.2	9d733441da13e683
Krazati (adagrasib) - FDA NDA 216340 multi-disciplinary review	288	538,880	9.2	b465c2a04d1defda
Inrebic (fedratinib) - FDA NDA 212327 multi-disciplinary review	257	513,288	15.2	012ee3ee2713bf50
Orgovyx (relugolix) - FDA NDA 214621 multi-disciplinary review	250	500,746	4.5	d696675b291cb536
Qinlock (ripretinib) - FDA NDA 213973 multi-disciplinary review	233	417,089	4.9	fc0a1d27a162b8fc
Nubeqa (darolutamide) - FDA NDA 212099 multi-disciplinary review	203	425,277	11.0	1c39d087378abb4c
Xpovio (selinexor) - FDA NDA 212306 multi-disciplinary review	188	342,190	6.2	b1a1803a8b22a070
Tazverik (tazemetostat) - FDA NDA 211723 multi-disciplinary review	189	412,998	11.6	68e62bfb7b5fb686
Exkivity (mobocertinib) - FDA NDA 215310 multi-disciplinary review	292	581,187	6.1	a65d2453dcf43ea3

Refused, and why that is a result

3 documents in the library cannot be searched, and each states its own reason rather than failing quietly. The refusals are load-bearing: `data/prep/split_review.py` refuses rather than trimming by hand, and a document that could be searched anyway would make that refusal decorative.

document	why it cannot be searched
TAK-994 (narcolepsy) - extracted findings, no source PDF	This case was assembled from extracted findings and has no source document in the repository, so there is nothing here to search. Its findings and their pages are on the case itself.
Tolcapone (Tasmar) - FDA medical review, 1998	48 of 48 pages carry almost no extractable text - this is a scanned document and needs OCR before anything can read it. REFUSED.
Troglitazone (Rezulin) - FDA approval package, 1997	No nonclinical chapter heading found after the contents. REFUSED - do not trim by hand.

3. Retrieval — did the answering page reach the model?

96.2%
hit@16
a page holding the answer was retrieved

91.5%
recall@16
of the pages that hold the answer

0.529
MRR
where the first gold page landed

33.7%
paraphrase stability
same question, different words, same pages

Measured over 53 answerable questions; 2 unanswerable questions are held aside, since they have no gold page by construction and counting them as misses would report a retrieval failure for a question the document cannot answer.

Where it started, and what moved it

The retriever was measured before it was changed, so every gain here is a difference between two runs of the same fixture rather than an assertion.

Where it started, and what moved it	hit@k	recall@k	MRR	stability	k	docs	n
BM25, as it was ✓	55.6%	50.0%	0.361	12.9%	8	3	18

+ stemming, phrase expansion, concept map ✓	83.3%	75.0%	0.546	35.7%	8	3	18
+ page boilerplate removed ✓	83.3%	77.8%	0.546	35.9%	8	3	18
+ eleven more documents ✓ Not an improvement and not meant as one - the retriever is unchanged here, and the row exists because every number above it was measured on three documents.	88.7%	79.2%	0.522	38.5%	8	14	53
+ sixteen pages retrieved instead of eight ✓	96.2%	**91.5%**	0.529	33.7%	16	14	53

Eight was never chosen against evidence - it was a reasonable default written before anything measured it. Sixteen is where the curve flattens; 24 scores better still and was declined, because every extra page is prose the model must not answer from.

How many pages an answer may read	hit@k	recall@k	MRR	stability	k	docs	n
k = 8	88.7%	79.2%	0.522	38.5%	8	14	53
k = 10	92.5%	87.7%	0.526	39.6%	10	14	53
k = 12	92.5%	87.7%	0.526	37.2%	12	14	53
k = 16 ✓	96.2%	91.5%	0.529	33.7%	16	14	53
k = 20	96.2%	91.5%	0.529	33.8%	20	14	53
k = 24	98.1%	93.4%	0.530	33.4%	24	14	53

Paraphrase stability by question group

Questions a reviewer would call the same question. This needs no answer key at all, which is why it is the most trustworthy number here: it began at 12.9%, meaning an acronym and its own expansion returned almost disjoint pages.

question group	page overlap between phrasings
inrebic:liver	39.1%
inrebic:noael	18.5%
krazati:liver	52.4%
krazati:noael	45.5%
lumakras:liver	39.1%
nipocalimab:margins	18.5%
nipocalimab:noael	26.6%
nipocalimab:reversibility	50.3%
orgovyx:liver	68.4%
orgovyx:noael	28.0%
qinlock:liver	45.5%
qinlock:noael	23.1%
retevmo:noael	6.7%

slynd:nonclinical	60.0%
tazverik:liver	45.5%
trikafta:noael	45.5%
turalio:clinical-reversibility	28.0%
turalio:liver-findings	16.1%
turalio:noael	10.3%
xpovio:noael	6.7%

4. Answers — what the model did with the pages

100.0% stated the required fact 95% CI 93.2%–100.0%, n=53	**100.0%** refused correctly on 2 questions the document cannot answer	**83.0%** citation recall of the pages nominated as gold	**0** errors upstream failures, truncations, refusals to parse

By document

An average over 14 documents can hide one that fails outright, so it is broken out. This is the varying-data question: does the number hold when the document changes?

document	questions	stated the fact	refused correctly
exkivity	1	1/1	–
inrebic	5	5/5	–
krazati	4	4/4	–
lumakras	3	3/3	–
nipocalimab	8	8/8	–
nubeqa	1	1/1	–
orgovyx	5	5/5	–
qinlock	4	4/4	–
retevmo	3	3/3	–
slynd	2	2/2	1/1
tazverik	3	3/3	–
trikafta	3	3/3	–
turalio	8	8/8	1/1
xpovio	3	3/3	–

5. What was built and rejected

Both of the obvious next upgrades were implemented, measured on this fixture, and removed. They are here because a report carrying only the improvements is an advertisement, and because the next person to have these ideas should see the numbers rather than repeat the work.

Sub-page windows

Scoring windows within a page rather than whole pages, so one relevant sentence is not diluted by two thousand characters of unrelated prose. Every setting that changed anything made recall worse, and the only setting that matched page-level scoring used a window larger than the median page - which is page scoring under another name.

Sub-page windows	hit@k	recall@k	MRR	stability	k	docs	n
whole page ✓	83.3%	**75.0%**	0.546	35.7%	8	3	18
60 / 30 tokens	83.3%	69.4%	0.581	30.1%	8	3	18
120 / 60 tokens	83.3%	66.7%	0.537	31.0%	8	3	18
240 / 120 tokens	77.8%	72.2%	0.583	31.8%	8	3	18
400 / 200 tokens	83.3%	75.0%	0.583	38.1%	8	3	18

Dense retrieval and hybrid fusion

All pages embedded with gemini-embedding-001 on Vertex, scored alone and fused with BM25 by reciprocal rank fusion. The best hybrid ties the lexical retriever on whether the answering page reaches the model at all, and buys only its position among the passages the model reads in full - against a network call per question, a vector cache larger than the documents, and the determinism that makes consistency attributable to the model rather than the retriever.

Dense retrieval and hybrid fusion	hit@k	recall@k	MRR	stability	k	docs	n
BM25 + term normalisation ✓	83.3%	**77.8%**	0.546	35.9%	8	3	18
dense only	83.3%	58.3%	0.395	37.3%	8	3	18
hybrid RRF 1:1	83.3%	69.4%	0.628	36.3%	8	3	18
hybrid RRF 2:1 lexical	83.3%	75.0%	0.644	34.2%	8	3	18

> **The failure was worth more than the feature.** Dense retrieval scored badly because every page begins with the same running header, so each page's vector encoded the document's identity rather than its content, with cosine scores bunched between 0.70 and 0.74. BM25 had hidden that for free, since idf discounts a term appearing everywhere. Removing the header lifted lexical recall and cut roughly 8% of the tokens from every prompt the surface sends.

6. Reproducing this

```
git checkout 6b626cf
npm ci
npm run retrieval:eval    # writes results/retrieval-eval.json
```

```
npm run ask:eval -- --repeats=3    # writes results/ask-eval.json, needs model credentials
npm run report    # writes this PDF from those two files
```

The approval-package PDFs are not committed — each is retrievable from accessdata.fda.gov by the NDA number in its filename, and the sha256 column above pins the bytes every number here was computed from.